

10.8 Solving Problems Involving Intersection of Secants

Lesson Introduction

 Benchmark	MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.			 Learning Target	I can solve problems involving secants drawn from a point external to the circle and their related angles.
 Benchmark Coherence	Building On			Working Toward	
	N/A			N/A	
 Essential Questions	What is the relationship between the angle formed by two secants intersecting in the exterior of a circle and intercepted arcs?				
 Lesson Narrative	In this lesson, students will learn about the angles formed by secants. Students will begin by exploring secants that intersect in the exterior of a circle to discover the exterior secant angle theorem. They will then apply this relationship to solve problems that involve algebraic expressions.				
 Component Summary	10.8.1 Warm-Up (~5 min.)	Reflecting on the importance of the ability to learn (<i>SE p. 160</i>)	10.8.3 Guided Practice (10-15 min.)	Solving problems involving angles formed by intersecting secants (<i>SE p. 161</i>)	
	10.8.2 Exploration (10 min.)	Discovering the relationship of angles formed by intersecting secants and their intercepted arcs (<i>SE p. 160</i>)	10.8.4 Your Turn! (15 min.)	Solving problems involving angles formed by intersecting secants (<i>SE p. 162</i>)	
	10.8.5 Wrap-Up (~5 min.)	Summarizing the lesson in a note to a classmate (<i>SE p. 163</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Secant - Exterior Secant Angle Theorem Additional Glossary: N/A		(Optional) GeoGebra Activity		Warm-Up 10.8.1 contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions • Students may reverse the subtraction of the arcs. • Students may apply central angle or inscribed angle relationships to secant angles.	Things to Consider • Be careful not to give anything away too early during the 10.8.2 Exploration, as it is designed to be a discovery of the Exterior Secant Angle Theorem.
	Literacy and Language Routines Think-Pair-Share During 10.8.2 Exploration, students will have independent time to think through and complete one section of the activity (a-c), and then collaborate with their partner in question 1 part d to compare conclusions before reviewing the formal definition of the exterior secant angle theorem.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure For 10.8.3 Guided Practice and 10.8.4 Your Turn!, students will be applying what they learned in the prior activity about angles formed by secants to solve mathematical problems that involve algebraic expressions. To do this, students will need to focus on the relevant details within the problem, in order to correctly find the difference between the intercepted arcs.
	 Differentiate Instruction Some learners may benefit from seeing how angles are formed by secants to better understand this conceptually. To do this, consider using digital applets, like GeoGebra’s “Angles formed by Secants” (outside).	
Lesson Notes:		

10.8.1 Warm-Up: Satellite Operations Engineer

Component Overview: Play the Career Profile video on the Satellite Operations Engineer for the entire class. After students watch the video (3:35 minutes), prompt them to reflect on the importance of being able to be trained by answering the one question independently.



10.8.1 Warm-Up: Satellite Operations Engineer

1. Veronica Navarro is a satellite operations engineer. Before beginning her career, she received a bachelor's degree. One piece of advice she would give to high school students is to take as many classes in high school as possible because they are free. In her opinion, a degree beyond high school is needed in many careers because employers want to know a person can be trained and is capable of learning. Veronica's employer paid for her to receive additional degrees as a means to further develop her abilities and deepen her learning.

Why do you think it is important for companies to know you are capable of being trained?

Answers will vary. Companies want to hire employees that will continue to grow as they work. They will continuously improve their abilities to perform the duties of their position and their contributions to the company will grow more substantial over time.



Allow students to share their responses. Students benefit when they hear the thinking of their peers.

10.8.2 Exploration: Angles Formed by Secants

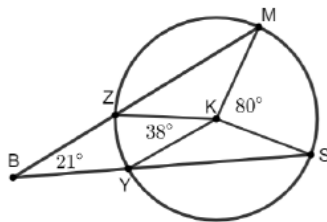
Component Overview: Students will explore angles formed by secants in the exterior of a circle through a four-part question series. For this activity, students will need a partner.



10.8.2 Exploration: Angles Formed by Secants

A **secant** is a line or line segment that passes through two points on a circle. In the interior of the circle, a secant line segment forms a chord.

1. $\angle MBS$ intersects Circle K , creating two intersecting arcs, where points M , Z , Y , and S are on the circle. The $m\angle ZKY = 38^\circ$, $m\angle SKM = 80^\circ$, and $m\angle MBS = 21^\circ$.



- a. Fill in the table.

Arc	Central Angle Measure	Arc Measure
$\widehat{MS}$	80°	80°
$\widehat{ZY}$	38°	38°

- b. Add and subtract the two arc measures.

$$m\widehat{MS} + m\widehat{ZY} = 118^\circ$$

$$m\widehat{MS} - m\widehat{ZY} = 42^\circ$$



This activity is designed to be an Exploration activity where students discover the Exterior Angle Theorem. Be careful not to give anything away too early in the activity when facilitating.



Watch student answers to question 1 part a. This can be an indicator of retention of the concepts from prior lessons. If students are stumped, refer them back to their notes from the central angle lesson earlier in the unit.

10.8.2 Exploration: Angles Formed by Secants (continued)

- c. Do the operations in Part B compare to the measure of $\angle MBS$? If so, explain.

The value of $m\angle MBS$ is half the difference of $\widehat{MS}$ and $\widehat{ZY}$.

- d. With your partner, write a formula that can be used to find the measure of an exterior angle of two secant lines.

$$m\angle MBS = \frac{1}{2}(\widehat{MS} - \widehat{ZY})$$



Exterior Secant Angle Theorem

The measure of an angle formed by two secants intersecting in the exterior of a circle is one-half the difference of the measures of the intercepted arcs.



Some learners may benefit from seeing how angles are formed by secants to better understand this conceptually. To do this, consider digital applets, like GeoGebra's "Angles formed by Secants" (outside).

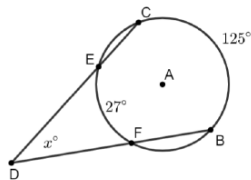
10.8.3 Guided Practice: Solving Problems Involving Angles Formed by Secants

Component Overview: Students will complete two structured practice questions where they will apply the relationship between angles formed by secants and their intercepted arcs to solve algebraic problems.



10.8.3 Guided Practice: Solving Problems Involving Angles Formed by Secants

- Circle A has points C , E , F , and B on the circle. Secant lines $\overline{CD}$ and $\overline{BD}$ intersect at point D outside the circle. The $m\widehat{CB} = 125^\circ$, $m\widehat{EF} = 27^\circ$, and $m\angle CDB = x^\circ$. What is the value of x ?



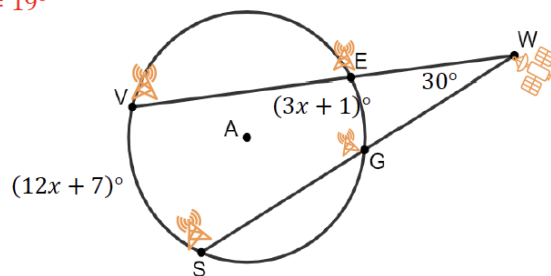
$$x = 49$$

- A satellite, point W , sends signals $\overline{VW}$ and $\overline{SW}$ to Earth. Signal $\overline{VW}$ hits tower E before reaching tower V . Similarly, signal $\overline{SW}$ hits tower G before reaching tower S . Scientists are interested in finding the measure of the intersecting arcs created by the two signals. They know $m\angle W = 30^\circ$, $m\widehat{VS} = (12x + 7)^\circ$, and $m\widehat{EG} = (3x + 1)^\circ$.

What is $m\widehat{VS}$ and $m\widehat{EG}$?

$$m\widehat{VS} = 79^\circ$$

$$m\widehat{EG} = 19^\circ$$



For this activity, consider beginning with a teacher-modeled activity for question 1. Students can benefit from hearing structured thinking out loud on determining what is known from the problem, what relationships can be determined, and how those relationships can be used to solve problems.



Question 2 provides a great call back to the career explored during the 10.8.1 Warm-Up. Consider highlighting this with the students to help debunk the myth that this is not applicable in the real-world.

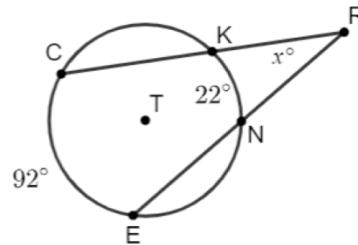
10.8.4 Your Turn!

Component Overview: Students will complete three practice questions where they will use their knowledge of the angle relationships between angles formed by secants and their intercepted arcs to solve algebraic problems.



10.8.4 Your Turn!

1. Circle T has points C , K , N , and E on the circle. Secant lines $\overline{CR}$ and $\overline{ER}$ intersect at point R outside the circle. The $m\widehat{CE} = 92^\circ$, $m\widehat{KN} = 22^\circ$, and $m\angle CRE = x^\circ$. What is the value of x ?

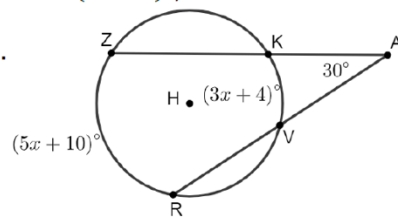


$$x = 35$$

2. Circle H has points Z , K , V , and R on the circle. Secant lines $\overline{ZA}$ and $\overline{RA}$ intersect at Point A outside the circle. The $m\widehat{ZR} = (5x + 10)^\circ$, $m\widehat{KV} = (3x + 4)^\circ$, and $m\angle ZAR = 30^\circ$.

Find the measure of $\widehat{ZR}$.

$$m\widehat{ZR} = 145^\circ$$



This activity is designed to give students independent practice on what they have learned in this lesson. Student performance on this activity can provide valuable insight into which parts students may be struggling with.



For students who finish early, have them create their own problem involving the relationship between intersecting secants and their intercepted arcs. Set the requirement that the measures of arcs and the angle must be written as algebraic expressions. Students should be cautious in creating a problem that, appropriately, has the closer arc measure smaller than the measure of the arc that is further from the angle.

10.8.5 Wrap-Up: Cool Down

Component Overview: Students will write a short note to a friend who missed class. This note will provide their own summary of what they learned today.



10.8.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed class summarizing today's lesson.

Answers will vary. Today we solved problems involving secants drawn from a point external to the circle. There were several real-world problems to help us figure out how to apply the concepts.



For students who might struggle representing their understanding, consider alternative ways to assess or measure learning for this lesson.

- If technology is available, students could record themselves explaining their summary.
- Students could provide detailed written methods of finding angles of intersecting chords.